

Session 1 — Foundation + Database

Build Phase 1 of MedFlow AI only.

Goal:

Create the foundation of a healthcare AI communication platform.

Infrastructure:

- App domain: <https://medflow.smartdeskai.cloud>
- n8n domain: <https://n8n.smartdeskai.cloud>
- Timezone: America/Chicago

Build only:

1. Next.js app structure
2. PostgreSQL + Prisma setup
3. Authentication structure
4. Role-based access control
5. Core dashboard layout
6. Core database models

Do NOT build AI calls, SMS, email, reminders, insurance, or medical records yet.

Core database models:

- User
- Role
- Client
- Appointment
- StaffTask
- StaffIntervention
- AuditLog
- ClientTimelineEvent
- ConsentRecord

Required appointment statuses:

SCHEDULED, CONFIRMED, RESCHEDULE_REQUESTED, RESCHEDULED, CANCELLED, CHECKED_IN, WAITING, WITH_PROVIDER, COMPLETED, NO_SHOW, FOLLOW_UP_NEEDED

Required staff intervention statuses:

WALK_IN, HUMAN TAKEOVER, AI_ESCALATED, STAFF_REVIEW_REQUIRED, RESOLVED, FOLLOW_UP_NEEDED, URGENT

Required roles:

ADMIN, MANAGER, FRONT_DESK_STAFF, BILLING_STAFF, MEDICAL_RECORDS_STAFF, CLINICAL_STAFF, READ_ONLY

Create dashboard pages:

- /dashboard
- /dashboard/clients
- /dashboard/appointments
- /dashboard/staff-intervention
- /dashboard/audit-logs
- /dashboard/settings

Every client must have a unified timeline.

Every important action must create an audit log.

Staff actions must always override AI automation.

Use clean, professional healthcare-friendly UI with Tailwind CSS.

Output:

- folder structure
- Prisma schema
- core pages
- API routes
- basic seed data
- README for Phase 1

Continue MedFlow AI from Phase 1.

Session 2 — Communication Logs

Build Phase 2 only: communication logging system.

Do NOT build real Twilio, Vapi, Gmail, or n8n integrations yet. Use service stubs/placeholders.

Add database models:

- CallLog
- SmsLog
- EmailLog
- AgentAction

Communication statuses:

PENDING, SENT, DELIVERED, ANSWERED, FAILED, NO_ANSWER, BOUNCED, RESPONDED, ESCALATED

Add dashboard pages:

- /dashboard/calls
- /dashboard/outbound-calls
- /dashboard/sms
- /dashboard/emails
- /dashboard/agent-coordination

Each call, SMS, and email must:

- connect to a client
- connect to an appointment when available
- create a ClientTimelineEvent
- create an AuditLog

Create service files:

- services/twilio.service.ts
- services/email.service.ts
- services/n8n.service.ts
- services/orchestrator.service.ts

For now, all services should be safe mock/stub implementations.

Add duplicate prevention:

Before creating or sending any communication, check if the same communication already exists for the same client, appointment, channel, and purpose.

Output:

- updated Prisma schema
- communication pages
- service stubs
- API routes
- README for Phase 2

Session 3 — Master Orchestrator + AI Agent Coordination

Continue MedFlow AI from Phase 2.

Build Phase 3 only: Master Orchestrator and AI agent coordination.

Do NOT build real calls, SMS, email, or reminders yet.

Create centralized orchestration logic.

Rules:

- AI agents must not act independently.

- Every AI action must go through the Master Orchestrator.
- No agent can directly modify appointments, clients, staff tasks, or communications without creating an AgentAction proposal.
- The Master Orchestrator approves, rejects, or escalates actions.
- Staff actions override AI actions.

Create agent definitions:

- Master Orchestrator Agent
- Front Desk AI Agent
- Intake AI Agent
- Appointment AI Agent
- Voice Call AI Agent
- SMS Reminder AI Agent
- Email AI Agent
- Staff Assistant AI Agent
- Escalation AI Agent

Each agent must have:

- system prompt
- allowed actions
- forbidden actions
- escalation rules
- human handoff rules

Create dashboard view:

- /dashboard/agent-coordination

Show:

- agent actions
- pending approvals
- approved actions
- rejected actions
- escalated actions

Add emergency safety flow:

If emergency language is detected:

- stop normal automation
- create urgent staff task
- create audit log
- mark case as urgent
- display in dashboard

Output:

- orchestrator service

- agent prompt files
- agent action workflow
- API routes
- dashboard page
- README for Phase 3

Session 4 — Walk-In + Physical Check-In

Continue MedFlow AI from Phase 3.

Build Phase 4 only: physical client management.

Create First-Time Walk-In Setup.

When a client physically arrives for the first time, staff must be able to:

- create client profile
- collect demographics
- collect phone/email/address
- collect preferred language
- collect communication preferences
- collect consent for voice, SMS, and email
- collect insurance information placeholder
- upload documents placeholder
- create first appointment
- create staff notes
- start onboarding workflow

Create Returning Client Physical Check-In.

When a returning client physically arrives, staff must be able to:

- search by name, phone, email, or appointment ID
- check client in
- update appointment status
- notify staff/provider
- create waiting room status
- log arrival time
- track wait duration
- mark WITH_PROVIDER, COMPLETED, WALK_OUT, or NO_SHOW

Add models:

- WalkInVisit
- PhysicalCheckIn
- WaitingRoomStatus

Add pages:

- /dashboard/walk-ins
- /dashboard/check-ins
- /dashboard/waiting-room

Every walk-in and check-in must:

- create ClientTimelineEvent
- create AuditLog
- notify Master Orchestrator
- respect staff override priority

Output:

- Prisma updates
- pages
- API routes
- timeline integration
- README for Phase 4

Session 5 — Appointment Reminder Engine

Continue MedFlow AI from Phase 4.

Build Phase 5 only: reminder engine logic.

Do NOT connect real Twilio, Vapi, Retell, Gmail, or n8n yet. Use service stubs.

Reminder system must be AI Voice First.

Every reminder cycle must use:

1. AI Voice Call
2. SMS
3. Email

Reminder schedule:

- 48 hours before appointment
- 24 hours before appointment
- 2 hours before appointment
- 30 minutes before appointment

Before sending any reminder:

- check consent
- check appointment status
- check duplicate reminder
- check if staff paused automation

Add model:

- ReminderLog

Reminder outcomes:

- CONFIRMED
- CANCELLED
- RESCHEDULE_REQUESTED
- NO_RESPONSE
- FAILED
- ESCALATED

If voice call fails:

- mark FAILED or NO_ANSWER
- send SMS
- send email
- create staff task for 2h and 30min failures

If client confirms:

- mark appointment CONFIRMED

If client cancels:

- mark appointment CANCELLED
- reopen slot
- notify staff

If client requests reschedule:

- mark RESCHEDULE_REQUESTED
- create staff task or offer available slots placeholder

Add page:

- /dashboard/reminders

Output:

- reminder engine service
- API routes
- dashboard page
- duplicate prevention
- consent checks
- README for Phase 5

Session 6 — Real Integrations

Continue MedFlow AI from Phase 5.

Build Phase 6 only: real external integrations.

Add .env.example keys:

```
OPENAI_API_KEY=  
TWILIO_ACCOUNT_SID=  
TWILIO_AUTH_TOKEN=  
TWILIO_PHONE_NUMBER=  
VAPI_API_KEY=  
VAPI_ASSISTANT_ID=  
RETELL_API_KEY=  
RETELL_AGENT_ID=  
GMAIL_CLIENT_ID=  
GMAIL_CLIENT_SECRET=  
GMAIL_REFRESH_TOKEN=  
GOOGLE_CLIENT_ID=  
GOOGLE_CLIENT_SECRET=  
GOOGLE_CALENDAR_ID=primary  
N8N_BASE_URL=https://n8n.smartdeskai.cloud  
N8N_WEBHOOK_INBOUND_CALL=  
N8N_WEBHOOK_OUTBOUND_CALL=  
N8N_WEBHOOK_SMS=  
N8N_WEBHOOK_EMAIL=  
N8N_WEBHOOK_APPOINTMENT=  
N8N_WEBHOOK_REMINDERS=  
N8N_WEBHOOK_STAFF_INTERVENTION=
```

Implement service integrations safely:

- Twilio Voice
- Twilio SMS
- Vapi or Retell AI
- Gmail or SendGrid
- Google Calendar
- n8n webhooks
- OpenAI

Important:

- all secrets must stay server-side
- never expose secrets in frontend
- validate all webhook requests
- add webhook secret validation
- keep mock mode available when env keys are missing

Output:

- updated services
- .env.example
- webhook API routes
- integration documentation
- README for Phase 6

Session 7 — n8n Workflows

Continue MedFlow AI from Phase 6.

Build Phase 7 only: n8n workflow JSONs and webhook documentation.

Generate importable n8n workflow JSONs for:

1. Inbound Call Handler
2. Outbound Call Handler
3. 48h Reminder Voice + SMS + Email
4. 24h Reminder Voice + SMS + Email
5. 2h Reminder Voice + SMS + Email
6. 30min Reminder Voice + SMS + Email
7. SMS Reply Handler
8. Email Reply Handler
9. Walk-In Onboarding Handler
10. Physical Check-In Handler
11. Human Handoff Handler
12. AI Coordination Handler

Use my n8n system:

<https://n8n.smartdeskai.cloud>

Each workflow must include:

- webhook trigger
- validation step
- call to MedFlow AI API
- error handler
- retry logic
- audit logging step
- clear notes

Output:

- /n8n-workflows/*.json
- webhook endpoint map

- setup guide
- README for Phase 7

Session 8 — HIPAA-Conscious Security Hardening

Continue MedFlow AI from Phase 7.

Build Phase 8 only: HIPAA-conscious security hardening.

Do not claim guaranteed HIPAA compliance.

Add documentation:

“Final HIPAA compliance requires legal review, vendor agreements, BAAs, organizational policies, staff training, and operational controls.”

Implement:

- PHI-safe logging
- RBAC enforcement
- audit logging enforcement
- session timeout
- webhook signature validation
- rate limiting
- secure headers
- data access checks
- export logging
- file access logging placeholder
- emergency escalation guardrails

AI safety rules:

- AI must not diagnose
- AI must not give treatment advice
- AI must identify itself as an AI assistant
- AI must escalate emergencies to staff and emergency services guidance

Output:

- security middleware
- audit middleware
- RBAC middleware
- security README
- HIPAA-conscious documentation

Session 9 — Deployment

Continue MedFlow AI from Phase 8.

Build Phase 9 only: deployment setup for my VPS.

Infrastructure:

- VPS uses Docker and Traefik
- App domain: <https://medflow.smartdeskai.cloud>
- n8n domain: <https://n8n.smartdeskai.cloud>

Generate:

- Dockerfile
- docker-compose.yml
- production .env.example
- Traefik labels
- deployment commands
- database migration commands
- backup instructions
- rollback instructions

The app must be ready to deploy to:
<https://medflow.smartdeskai.cloud>

Output:

- deployment files
- production checklist
- README for Phase 9